

# Jake Russin

jake\_russin@brown.edu

Personal website

Google Scholar

## CURRENT APPOINTMENT

---

|                                                                                                                    |                |
|--------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Brown University</b><br>Postdoctoral Research Associate<br>Advisors: Dr. Michael J. Frank and Dr. Ellie Pavlick | 2023 – present |
|--------------------------------------------------------------------------------------------------------------------|----------------|

## EDUCATION

---

|                                                                                                                                                                                                              |             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>PhD in Psychology, University of California, Davis</b><br>Thesis: <i>Compositionality and Cognitive Control in Neural Networks</i><br>Advisor: Dr. Randall O'Reilly (transferred from CU Boulder in 2019) | 2017 – 2023 |
| <b>Non-Degree Students Program, Yale University</b><br>Full course load in statistics, mathematics, machine learning                                                                                         | 2016 – 2017 |
| <b>BA in Neuroscience and Philosophy, Colorado College</b><br>Thesis: <i>Reasoning in the Prefrontal Cortex</i>                                                                                              | 2010 – 2014 |

## OTHER EMPLOYMENT

---

|                                                                                                                                              |                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Meta AI</b><br><i>Research Intern, Advisors: Dr. Brenden Lake and Dr. Dieuwke Hupkes</i>                                                  | Summer 2022<br>New York, NY          |
| <b>Microsoft Research</b><br><i>Research Intern, Advisors: Dr. Paul Smolensky and Roland Fernandez</i>                                       | Summer 2020<br>Redmond, WA (virtual) |
| <b>Montreal Institute for Learning Algorithms (MILA)</b><br><i>Visiting Student Researcher, Advisors: Dr. Yoshua Bengio and Dr. Jason Jo</i> | Summer 2018<br>Montreal, Canada      |
| <b>National Institutes of Health</b><br><i>Postbac Intramural Research Training Award, Advisor: Dr. Carolyn Beebe Smith</i>                  | 2015 – 2016<br>Bethesda, MD          |
| <b>Institute of Cognitive Science, University of Colorado, Boulder</b><br><i>Research Assistant, Advisor: Dr. Albert Kim</i>                 | 2014 – 2015<br>Boulder, CO           |

## SELECTED AWARDS AND GRANTS

---

|                                                                                                                       |              |
|-----------------------------------------------------------------------------------------------------------------------|--------------|
| T32 Training program fellowship in computational psychiatry<br><i>National Institute of Mental Health</i>             | 2025-present |
| Summer Seminars in Neuroscience and Philosophy Fellow (\$17,720)<br><i>Duke University, John Templeton Foundation</i> | 2022-2023    |
| T32 Training program fellowship in learning, memory, and plasticity<br><i>National Institute of Mental Health</i>     | 2020-2021    |
| NSF Graduate Research Fellowships Program honorable mention<br><i>National Science Foundation</i>                     | 2018         |
| Postbac Intramural Research Training Award<br><i>National Institutes of Health</i>                                    | 2015-2016    |
| Phi Beta Kappa Society                                                                                                | 2014         |

|                                                                                 |            |
|---------------------------------------------------------------------------------|------------|
| The J. Glenn Gray Award in Philosophy, <i>Colorado College</i>                  | 2014       |
| Distinction in Neuroscience, Distinction in Philosophy, <i>Colorado College</i> | 2014       |
| Dean's List, junior and senior years, <i>Colorado College</i>                   | 2013, 2014 |
| Psi Chi Honor Society                                                           | 2013       |

## PUBLICATIONS

---

### AVAILABLE PREPRINTS

**Russin, J.\***, McGrath, S. W.\*, Williams, D. (2024) From Frege to ChatGPT: Compositionality in language, cognition, and deep neural networks. Forthcoming in *Neuroscience and Philosophy II* [arXiv]

Park S., Zolfaghar M.\*, **Russin, J.\***, Boorman, E., O'Reilly, R. C. (2022) The representational geometry of cognitive maps under dynamic cognitive control. *Under review at Nature Communications*. [bioRxiv]

### PEER-REVIEWED JOURNAL ARTICLES AND CONFERENCE PAPERS

**Russin, J.**, Pavlick, E., Frank, M. J. (2025) Parallel trade-offs in human cognition and neural networks: The dynamic interplay between in-context and in-weight learning. *Proceedings of the National Academy of Sciences*, 125(35). [PNAS]

**Russin, J.\***, Kim, J.\*, Pavlick, E., Frank, M. J. (2025) Step-by-step analogical reasoning in humans and neural networks. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*. [CogSci]

McGrath, S. W., **Russin, J.** (2024) Multiple realizability and the rise of deep learning. *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*. [CogSci]

**Russin, J.**, McGrath, S. W., Pavlick, E., Frank, M. J. (2024) Is human compositionality meta-learned? *Commentary in Behavioral and Brain Sciences*. Target article: *Meta-Learned Models Cognition*. [BBS]

Webb T. W., Frankland, S. M., Altabaa, A., Krishnamurthy, K., Campbell, D., **Russin, J.**, O'Reilly, R., Lafferty, J., Cohen, J. D. The relational bottleneck as an inductive bias for efficient abstraction. *Trends in Cognitive Sciences*. [TICS]

McGrath, S. W.\*, **Russin, J.\***, Pavlick, E., Feiman, R. (2024) How can deep neural networks inform theory in psychological science? *Current Directions in Psychological Sciences*. [CDPS]

McGrath, S. W.\*, **Russin, J.\***, Pavlick, E., Feiman, R. (2023) Properties of LoTs: The footprints or the bear itself? *Commentary in Behavioral and Brain Sciences*. Target article: *The Best Game in Town: The Re-Emergence of the Language of Thought Hypothesis Across the Cognitive Sciences*. [BBS]

Chakravarthy, A.\*, **Russin, J.\***, O'Reilly, R. C. (2022) Systematicity emerges in transformers when abstract grammatical roles guide attention. *Proceedings of the 2022 Conference of the NA Chapter of the Association for Computational Linguistics, Student Research Workshop*. [NAACL SRW]

Zolfaghar M.\*, **Russin, J.\***, Park S.\*, Boorman, E., O'Reilly, R. C. (2022) The geometry of map-like representations under dynamic cognitive control. *Proceedings of the 44th Annual Meeting of the Cognitive Science Society*. [CogSci]

**Russin, J.**, Zolfaghar M., Park S., Boorman, E., O'Reilly, R. C. (2022) A neural network model of continual learning with cognitive control. *Proceedings of the 44th Annual Meeting of the Cognitive Science Society*. [CogSci]

O'Reilly, R. C., Ranganath, C., **Russin, J.** (2022) The structure of systematicity in the brain. *Current Directions in Psychological Science*. [CDPS]

**Russin, J.**, O'Reilly, R. C. (2021) Deep predictive learning with local gradient information. *Abstract accepted for poster presentation at Bernstein Conference 2021*. [Bernstein]

---

\*Equal contribution

**Russin, J.**, Fernandez, R., Palangi, H., Rosen, E., Jojic, N., Smolensky, P., Gao, J. (2021). Compositional processing emerges in neural networks solving math problems. *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society*. [CogSci]

**Russin, J.\***, Zolfaghar, M.\*, Park, S. A., Boorman, E., O'Reilly, R. C. (2021). Complementary structure-learning neural networks for relational reasoning. *Proceedings of the 43rd Annual Meeting of the Cognitive Science Society*. [CogSci]

O'Reilly, R. C., **Russin, J.**, Zolfaghar, M., Rohrlich, J. (2021). Deep predictive learning in neocortex and pulvinar. *Journal of Cognitive Neuroscience*, 33(6), 1158–1196. [JCN]

**Russin, J.**, O'Reilly, R. C., Bengio, Y. (2020). Deep learning needs a prefrontal cortex. *Bridging AI and Cognitive Science (BAICS) Workshop, ICLR 2020, 11*. [BAICS]

**Russin, J.**, Jo, J., O'Reilly, R. C., Bengio, Y. (2020). Systematicity in a recurrent neural network by factorizing syntax and semantics. *Proceedings of the 42nd Annual Meeting of the Cognitive Science Society, 7*. [CogSci]

O'Reilly, R. C., Nair, A., **Russin, J.**, Herd, S. A. (2020). How sequential interactive processing within frontostriatal loops supports a continuum of habitual to controlled processing. *Frontiers in Psychology*, 11. [frontiers]

Cooke, S. K., **Russin, J.**, Moulton, K., Nadel, J., Loutaev, I., Gu, Q., Li, Z., Smith, C. B. (2019). Effects of the presence and absence of amino acids on translation, signaling, and long-term depression in hippocampal slices from Fmr1 knockout mice. *Journal of Neurochemistry*. [JNC]

## BOOK CHAPTERS

O'Reilly, R. C., **Russin, J.**, Herd, S. A. (2019). Computational models of motivated frontal function. *Handbook of Clinical Neurology*, 163, 317–332. [ScienceDirect]

## IN PREPARATION

**Russin, J.**, Pavlick, E., Frank, M. J. (*in prep*) Transformers capture individual differences in reinforcement learning via in-context learning.

## CONFERENCE PRESENTATIONS

---

**Russin, J.**, Kim, J. (2025, July) Step-by-step analogical reasoning in humans and neural networks. Poster presented at: CogSci 2025, San Francisco, CA.

**Russin, J.**, Kim, J. (2025, June) Step-by-step analogical reasoning in humans and neural networks. Poster presented at: RLDM 2025, Dublin, Ireland.

**Russin, J.** (2025, March) Compositionality in neural networks capable of in-context learning. Talk presented at: Towards an Algorithmic Framework for Studying Compositionality in Biological and Artificial Neural Networks, Cosyne 2025 workshop, Mont-Tremblant, Quebec, CA.

**Russin, J.** (2024, August) Emergent Cognitive Flexibility in Large Language Models. Talk presented virtually at: Robophilosophy 2024, Aarhus University.

**Russin, J.** (2024, August) From Frege to chatGPT: Compositionality in language, cognition, and deep neural networks. Talk presented at: Summer Seminars in Neuroscience and Philosophy, Duke.

**Russin, J.** (2024, August) Curriculum Effects and Compositionality Emerge with In-Context Learning in Neural Networks. Poster presented at: Cognitive Computational Neuroscience conference, MIT.

**Russin, J.** and McGrath, S. (2024, July) Compositionality in Large Language Models. Talk presented at: Descartes Lectures Workshops 2024, Tilburg, the Netherlands.

- McGrath, S. and **Russin, J.** (2024, July) Multiple Realizability and the Rise of Deep Learning. Poster presented at: CogSci 2024, Rotterdam, the Netherlands.
- Russin, J.** (2024, July) Human Curriculum Effects Emerge with In-Context Learning in Neural Networks. Talk presented at: CogSci 2024, Rotterdam, the Netherlands.
- Russin, J.** (2024, May) Human Curriculum Effects Emerge with In-Context Learning in Neural Networks. Talk presented at: Control Processes 2024, Birmingham, UK.
- Russin, J.** and McGrath, S. (2023, June). Can Deep Learning Inform Explanation in Cognitive Science? Talk presented at: SPP 2023, Pittsburgh, PA.
- Russin, J.** (2022, July). The Geometry of Map-Like Representations under Dynamic Cognitive Control. Talk presented at: CogSci 2022, Toronto, CA.
- Russin, J.** (2022, July). A Neural Network Model of Continual Learning with Cognitive Control. Talk presented at: CogSci 2022, Toronto, CA.
- Russin, J.** (2022, July). Systematicity Emerges in Transformers when Abstract Grammatical Roles Guide Attention. Poster presented at: NAACL SRW 2022, Seattle, WA.
- Russin, J.** (2022, June). A Neural Network Model of Continual Learning with Cognitive Control. Poster presented at: RLDM 2022, Providence, RI.
- Russin, J.** (2022, March). The Geometry of Map-Like Representations under Dynamic Cognitive Control. Poster presented at: COSYNE 2022, Lisbon, Portugal.
- Russin, J.** (2021, September). Deep predictive learning with local gradient information. Poster presented at: Bernstein Conference 2021.
- Russin, J.** (2021, July). Complementary structure-learning neural networks for relational reasoning. Poster presented at: 43rd Annual Meeting of the Cognitive Science Society.
- Russin, J.** (2021, July). Compositional processing emerges in neural networks solving math problems. Poster presented at: 43rd Annual Meeting of the Cognitive Science Society.
- Russin, J.** (2020, April) Deep learning needs a prefrontal cortex. Talk presented at: Bridging AI and Cognitive Science, ICLR 2020 Workshop, Addis Ababa, Ethiopia, April 26, 2020.
- Russin, J.** (2020, July). Systematicity in a recurrent neural network by factorizing syntax and semantics. Talk presented at: 42nd Annual Meeting of the Cognitive Science Society.
- Russin, J.** (2020, July). Compositional generalization by factorizing alignment and translation. Talk presented at: ACL 2020, Student Research Workshop, Seattle, WA.

## INVITED TALKS

---

- |         |                                                                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/2025 | <b>Russin, J.</b> Parallel trade-offs in human cognition and neural networks: The dynamic interplay between in-context and in-weight learning. MindRL Hub.              |
| 05/2025 | <b>Russin, J.</b> The dynamic interplay between in-context and in-weight learning in humans and neural networks. Schuck lab, University of Hamburg.                     |
| 04/2022 | <b>Russin, J.</b> Continual learning and map-like representations in neural networks. Human information Processing lab (Summerfield), Oxford University, Oxford, UK.    |
| 04/2022 | <b>Russin, J.</b> Map-like representations and relational reasoning in neural networks. The Reasoning Lab (Khemlani), Naval Research Laboratory, Washington DC          |
| 07/2021 | <b>Russin, J.</b> Structure-learning and compositionality in neural networks. Laboratory of Neural Computation and Cognition (Frank), Brown University, Providence, RI. |

|         |                                                                                                                                                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03/2021 | <b>Russin, J.</b> Complementary structure-learning systems. PCCN Brownbag, Department of Psychology, University of California Davis, Davis, CA.                           |
| 05/2020 | <b>Russin, J.</b> Systematicity in neural networks. PCCN Brownbag, Department of Psychology, University of California Davis, Davis, CA.                                   |
| 04/2020 | <b>Russin, J.</b> Generalization, structure, and cognitive maps. Memory Meeting, Center for Neuroscience, University of California Davis, Davis, CA.                      |
| 09/2019 | <b>Russin, J.</b> Temporal abstraction in reinforcement learning. Learning and Decision-Making Lab, Center for Mind and Brain, University of California Davis, Davis, CA. |
| 05/2019 | <b>Russin, J.</b> Systematicity, selective attention, and the prefrontal cortex. 38th Annual Ekstrand Mini-Convention, University of Colorado, Boulder, Boulder, CO.      |
| 12/2018 | <b>Russin, J.</b> Compositional Attention Networks. Statistics, Optimization, and Machine Learning Seminar, University of Colorado, Boulder, Boulder, CO.                 |
| 10/2018 | <b>Russin, J.</b> Temporal abstraction in reinforcement learning. Kilpatrick Group, Department of Applied Mathematics, University of Colorado, Boulder, Boulder, CO.      |
| 05/2018 | <b>Russin, J.</b> Temporal abstraction in reinforcement learning. 37th Annual Ekstrand Mini-Convention, University of Colorado, Boulder, Boulder, CO.                     |
| 05/2014 | <b>Russin, J.</b> Reasoning in the prefrontal cortex. Psychology and Neuroscience Poster Day, Colorado College, Colorado Springs, CO.                                     |

## TEACHING

---

|              |                                                                                                                                                                  |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9/24-present | SOLAR undergraduate research club, Brown University<br><i>Help lead weekly meeting for LUNAR lab undergraduates to present research.</i>                         |
| Jan. 2025    | Primer on large language models for cognitive scientists, Brown University.<br><i>Lectured on LLMs for Cognitive and Psychological Sciences Retreat 2025</i>     |
| Spring 2018  | Teaching assistant: Research Methods in Psychology, CU Boulder.<br><i>Lectured and led discussions in weekly lab sections, graded, held weekly office hours.</i> |
| Fall 2017    | Teaching assistant: History of Psychology, CU Boulder.<br><i>Graded, attended lectures, held weekly office hours.</i>                                            |

## MENTORING

---

|                |                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 – present | Tianze (Ethi) Hua (PhD student, Computer Science, Brown)<br><i>Research mentor for projects on mechanistic interpretability in LLMs</i>                 |
| 2024 – present | Joonhwa Kim (PhD student, Neuroscience, Brown)<br><i>Research mentor for projects involving analogical reasoning in humans and LLMs</i>                 |
| 2023 – present | Krishn Bera (PhD student, Cognitive and Psychological Sciences, Brown)<br><i>Research mentor for projects involving LLMs and cognitive control</i>      |
| 2021 – 2023    | Ayush Chakravarthy (undergraduate, Computer Science, UC Davis)<br><i>Research mentor for projects in deep learning and natural language processing.</i> |
| 2020 – 2022    | Andrew Carlson (PhD student, Psychology, UC Davis)<br><i>Research mentor and advisor on career development, publishing, presentations, etc.</i>         |
| 2021           | Graduate student mentor for CogSci 2021 Mentoring Program                                                                                               |

2020 – 2021      Taiqi He (junior specialist postbac, UC Davis)  
*Research mentor for projects in deep learning and natural language processing.*

## SERVICE

---

2019 – present      Ad hoc reviewer: *PNAS*, *Nature Neuroscience*, *Nature Human Behavior*, *Nature Communications Biology*, *TICS*, *Current Opinion in Behavioral Sciences*, *Psychonomic Bulletin and Review*

2020 – present      Conference reviewer: CogSci conference (2020, 2021, 2022, 2023, 2024, 2025), CCN conference (2024), RLDM conference (2025)

2026                  Workshop organizer: “Mechanistic interpretability in brains and machines: shared concepts and converging methods” COSYNE 2026

2023 – 2024          Carney Brain Science External Postdoc Seminar, Speaker Selection Committee

2022 – 2024          Reading group on AI in philosophy and cognitive science: led weekly zoom meeting for researchers at multiple institutions (UC Davis, Brown, UPenn, Stanford)